

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Text;
5 using System.Threading.Tasks;
6 using System.Text.Json;
7 using System.Text.Json.Serialization;
8 using CustomProjectRPG.Interfaces;
9 using CustomProjectRPG.Entities.Enum;
10 using CustomProjectRPG.Entities.Component;
11 using CustomProjectRPG.Core.Utilities;
12
13 namespace CustomProjectRPG.Entities
14 {
15
16
17
18 public class BeastEntity : ISerializeJSON<BeastEntity>
19     {
20         private CustomProjectRPG.Entities.Enum.Enum.BeastType _beastType;
21         private CorruptionComponent _corruption;
22         private StatsComponent _stats;
23         private bool _isAggressive;
24
25         public event Action<string, CustomProjectRPG.Entities.Enum.Enum.CorruptionState>
26             CorruptionStateChanged; // Event for state updates
27
28         /// <summary>
29         /// Initializes a new BeastEntity with the specified beast type
30         /// and initial stats.
31         /// </summary>
32         public BeastEntity(CustomProjectRPG.Entities.Enum.Enum.BeastType
33             beastType)
34         {
35             _beastType = beastType;
36             _corruption = new CorruptionComponent
37                 (CustomProjectRPG.Entities.Enum.Enum.EntityType.Beast,
38                 GetInitialCorruption(beastType));
39             _stats = new StatsComponent
40                 (CustomProjectRPG.Entities.Enum.Enum.EntityType.Beast,
41                 GetInitialHP(), GetInitialATK(), GetInitialDEF(),
42                 GetInitialAgility());
43             _isAggressive = false;
44             UpdateAggressiveness();
45         }
46
47         /// <summary>
48         /// Gets the initial corruption percentage based on beast type.
```

```
41     /// </summary>
42     private float GetInitialCorruption
43     (CustomProjectRPG.Entities.Enum.Enum.BeastType beastType)
44     {
45         return beastType switch
46         {
47             CustomProjectRPG.Entities.Enum.Enum.BeastType.Weak => 0f,
48             CustomProjectRPG.Entities.Enum.Enum.BeastType.Elite => 50f,
49             CustomProjectRPG.Entities.Enum.Enum.BeastType.Boss => 75f,
50             _ => 0f
51         };
52     }
53     /// <summary>
54     /// Gets the initial HP based on beast type.
55     /// </summary>
56     private float GetInitialHP()
57     {
58         return _beastType switch
59         {
60             CustomProjectRPG.Entities.Enum.Enum.BeastType.Weak => 50f,
61             CustomProjectRPG.Entities.Enum.Enum.BeastType.Elite => 150f,
62             CustomProjectRPG.Entities.Enum.Enum.BeastType.Boss => 300f,
63             _ => 50f
64         };
65     }
66     /// <summary>
67     /// Gets the initial ATK based on beast type.
68     /// </summary>
69     private float GetInitialATK()
70     {
71         return _beastType switch
72         {
73             CustomProjectRPG.Entities.Enum.Enum.BeastType.Weak => 5f,
74             CustomProjectRPG.Entities.Enum.Enum.BeastType.Elite => 15f,
75             CustomProjectRPG.Entities.Enum.Enum.BeastType.Boss => 25f,
76             _ => 5f
77         };
78     }
79     /// <summary>
80     /// Gets the initial DEF based on beast type.
81     /// </summary>
82     private float GetInitialDEF()
```

```
85     {
86         return _beastType switch
87         {
88             CustomProjectRPG.Entities.Enum.Enum.BeastType.Weak => 2f,
89             CustomProjectRPG.Entities.Enum.Enum.BeastType.Elite => 8f,
90             CustomProjectRPG.Entities.Enum.Enum.BeastType.Boss => 15f,
91             _ => 2f
92         };
93     }
94
95     /// <summary>
96     /// Gets the initial Agility based on beast type.
97     /// </summary>
98     private float GetInitialAgility()
99     {
100         return _beastType switch
101         {
102             CustomProjectRPG.Entities.Enum.Enum.BeastType.Weak => 3f,
103             CustomProjectRPG.Entities.Enum.Enum.BeastType.Elite => 10f,
104             CustomProjectRPG.Entities.Enum.Enum.BeastType.Boss => 7f,
105             _ => 3f
106         };
107     }
108
109     /// <summary>
110     /// Updates the beast's aggressiveness based on corruption state.
111     /// </summary>
112     private void UpdateAggressiveness()
113     {
114         var state = _corruption.GetCorruptionState();
115         _isAggressive = state == CustomProjectRPG.Entities.Enum.Enum.CorruptionState.Full;
116         CorruptionStateChanged?.Invoke($"Beast_{_beastType}", state); // Notify state change
117     }
118
119     /// <summary>
120     /// Applies damage to the beast, updating stats and checking death.
121     /// </summary>
122     public void TakeDamage(float amount)
123     {
124         _stats.TakeDamage(amount);
125         if (_stats.HP <= 0)
126         {
127             // Handle death logic (e.g., drop items, quest update)
128             Console.WriteLine($"Beast {_beastType} defeated!");
129         }
130     }
131 }
```

```
130     }
131
132     /// <summary>
133     /// Increases corruption, triggering state and aggressiveness updates.
134     /// </summary>
135     public void IncreaseCorruption(float amount)
136     {
137         _corruption.IncreaseCorruption(amount);
138         UpdateAggressiveness();
139     }
140
141     /// <summary>
142     /// Serializes the beast's state to JSON.
143     /// </summary>
144     public string Serialize()
145     {
146         var data = new BeastData
147         {
148             BeastType = _beastType,
149             CorruptionState = _corruption.Serialize(),
150             StatsState = _stats.Serialize(),
151             IsAggressive = _isAggressive
152         };
153         return JsonSerializer.Serialize(data);
154     }
155
156     /// <summary>
157     /// Deserializes the beast's state from JSON.
158     /// </summary>
159     public void Deserialize(string json)
160     {
161         var data = JsonSerializer.Deserialize<BeastData>(json)
162             ?? throw new JsonException("Invalid JSON format for beast state.");
163         _beastType = data.BeastType;
164         _corruption.Deserialize(data.CorruptionState);
165         _stats.Deserialize(data.StatsState);
166         _isAggressive = data.IsAggressive;
167         UpdateAggressiveness();
168     }
169
170     private class BeastData
171     {
172         [JsonPropertyName("beastType")]
173         public CustomProjectRPG.Entities.Enum.Enum.BeastType
174             BeastType { get; set; }
175
176         [JsonPropertyName("corruptionState")]
```

```
176         public string CorruptionState { get; set; }
177
178         [JsonPropertyName("statsState")]
179         public string StatsState { get; set; }
180
181         [JsonPropertyName("isAggressive")]
182         public bool IsAggressive { get; set; }
183     }
184 }
185 }
186
187
```